



DP-600 : Implement analytics solutions using Microsoft Fabric

Exercise 8 : Transform data with T-SQL in a Fabric warehouse

- **Create a workspace**
- **Create a warehouse**
- **Transform data with T-SQL queries**
- **Create a view for reusable logic**
- **Build a stored procedure with parameters**
- **Create and load dimensional tables**
- **Verify the final state**



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Create a workspace

Create a workspace

Name *
Transform_data_tsq_8
✓ This name is available

Description
Describe this workspace

Domain ⓘ
Assign to a domain (optional)

[Learn more about workspace settings](#)

Workspace image
Upload
Reset

Advanced ^

Contact list * ⓘ
rsidvika (Owner)

Workspace type ⓘ
What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#)

Power BI only workspace types
Power BI workspaces only support the use of Power BI capabilities.

Apply Cancel

WS name: Transform_data_tsq_8

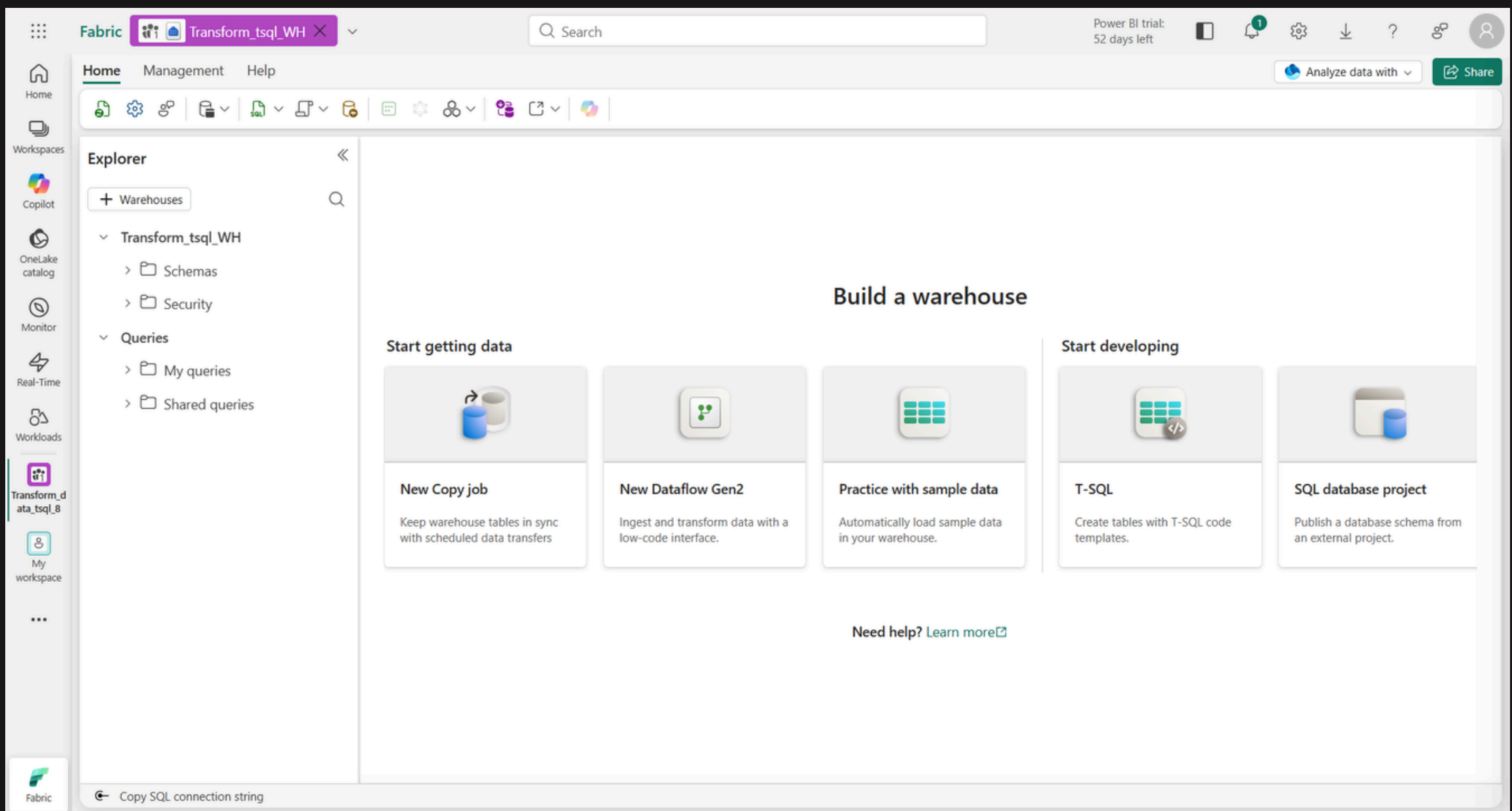
WS Type: Trial



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Create a warehouse



LH name: Transform_tsqI_WH



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► Transform data with T-SQL queries

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SQL query 2

Run

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Create rule

Explain query

Fix query errors

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-- Filter completed orders and add calculated columns

SELECT

order_id,

customer_id,

order_date,

quantity,

unit_price,

quantity * unit_price AS line_total,

ISNULL(discount,0) AS Discount,

(quantity * unit_price)-ISNULL(discount,0) AS net_amount,

CASE

WHEN quantity*unit_price > 200 THEN 'High'

WHEN quantity*unit_price > 100 THEN 'Medium'

ELSE 'Standard'

END AS order_tier

FROM staging.orders

where status='Completed';

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Results



DP-600 : Implement analytics solutions using Microsoft Fabric

► Transform data with T-SQL queries

SQL query 2

Run Save as Save as view Create rule Explain query Fix query errors

```
1
2 select c.region,c.segment, count(*) AS order_Count,
3       count(*) as order_count,
4       SUM(o.quantity*o.unit_price) AS total_sales,
5       AVG(o.quantity*o.unit_price) as avg_order_value
6 from staging.orders as o
7 INNER JOIN staging.customers AS c
8       ON o.customer_id=c.customer_id
9 where o.status='Completed'
10 GROUP BY c.region, c.segment
11 ORDER BY total_sales desc;
```

Results						
	ABC region	ABC segment	123 order_Count	123 order_count	e* total_sales	e* avg_order_value
1	West	SMB	3	3	1074.60	358.200000
2	West	Enterprise	2	2	675.00	337.500000
3	East	Enterprise	3	3	645.00	215.000000
4	North	Enterprise	3	3	645.00	215.000000



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Create a view for reusable logic

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SQL query 2

Run

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Save as view

Create rule

Explain query

Fix query errors

```
1
2
3 create view gold.vw_monthly_sales
4 as
5 select
6     d.calendar_year,
7     d.calendar_month,
8     d.month_name,
9     p.category,
10    count(*) as order_count,
11    SUM(o.quantity) as total_quantity,
12    sum(o.quantity* o.unit_price) as total_sales
13 from staging.orders as o
14 inner join staging.dates d
15     on o.order_date=d.calendar_date
16 inner join staging.products as p
17     on o.product_id=p.product_id
18 where o.status='Completed'
19 GROUP by d.calendar_year, d.calendar_month, d.month_name,p.category;
20
```

Messages

Copy messages

6:45:37 PM Started executing on line 1

6:45:37 PM Query execution time: 00:00:00.014 | Time to process and render results: 00:00:00.631 | Total duration: 00:00:00.645

Fabric

AutoSave: On (Saved) Succeeded (0 sec 645 ms) Copilot completions: Off

Results							
	123 calendar_year	123 calendar_month	ABC month_name	ABC category	123 order_count	123 total_quantity	e ^x total_sales
1	2026	1	January	Gadgets	1	5	225.00
2	2026	1	January	Widgets	1	10	150.00
3	2026	2	February	Gadgets	1	3	180.00
4	2026	2	February	Widgets	1	20	450.00
5	2026	3	March	Widgets	1	15	225.00
6	2026	4	April	Widgets	2	37	645.00
7	2026	5	May	Components	1	40	399.60
8	2026	5	May	Gadgets	1	6	360.00
9	2026	6	June	Gadgets	1	2	90.00
10	2026	6	June	Widgets	1	14	315.00



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Build a stored procedure with parameters

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Fabric

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SQL query 3

SQL query 4

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-- Create stored procedure: refresh monthly sales for a given year and month

CREATE PROCEDURE gold.usp_refresh_monthly_sales

@year INT,

@month INT

AS

BEGIN

-- Remove existing data for the target period

DELETE FROM gold.monthly_sales

WHERE calendar_year = @year AND calendar_month = @month;

-- Insert fresh aggregated data

INSERT INTO gold.monthly_sales

(calendar_year, calendar_month, month_name, category,

order_count, total_quantity, total_sales)

SELECT

d.calendar_year,

d.calendar_month,

d.month_name,

p.category,

COUNT(*),

SUM(o.quantity),

SUM(o.quantity * o.unit_price)

FROM staging.orders AS o

INNER JOIN staging.dates AS d

ON o.order_date = d.calendar_date

INNER JOIN staging.products AS p

ON o.product_id = p.product_id

Messages

Copy messages

AutoSave: On (Last saved: 3m ago)

Succeeded (0 sec 764 ms)

Copilot completions: Off



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stored procedure with parameters Execution

Workspaces

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Monitor

SQL query 2SQL query 3SQL query 4

RunSave asSave as viewCreate ruleExplain queryFix query errors

1

-- Execute the procedure for January 2026 and view results

2

EXEC gold.usp_refresh_monthly_sales @year = 2026, @month = 1;

3

4

SELECT * FROM gold.monthly_sales;

	123	calendar_year	123	calendar_month	ABC	month_name	ABC	category	123	order_count	123	total_quantity	e*	total_sales
1		2026		1		January		Gadgets	1			5		225.00
2		2026		1		January		Widgets	1			10		150.00



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► Create and load dimensional tables

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Transform data to SQL

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SQL query 5

SQL query 6

SQL query 7

SQL query 8

Run

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Save as view

Create rule

Explain query

Fix query errors

```
1  -- Verify: query fact table joined back to dimensions
2  SELECT
3      d.calendar_date,
4      d.month_name,
5      c.customer_name,
6      c.segment,
7      p.product_name,
8      p.category,
9      f.quantity,
10     f.sales_amount
11 FROM fact.sales AS f
12 INNER JOIN dim.date AS d ON f.date_key = d.date_key
13 INNER JOIN dim.customer AS c ON f.customer_key = c.customer_key
14 INNER JOIN dim.product AS p ON f.product_key = p.product_key
15 ORDER BY d.calendar_date;
```

Results								
	calendar_date	month_name	customer_name	segment	product_name	category	quantity	sales_amount
1	2026-01-15	January	Contoso Ltd	Enterprise	Widget Alpha	Widgets	10	150.00
2	2026-01-22	January	Contoso Ltd	Enterprise	Gadget Gamma	Gadgets	5	225.00
3	2026-02-10	February	Fabrikam Inc	SMB	Widget Beta	Widgets	20	450.00
4	2026-02-18	February	Northwind Traders	Enterprise	Gadget Delta	Gadgets	3	180.00
5	2026-03-05	March	Fabrikam Inc	SMB	Widget Alpha	Widgets	15	225.00
6	2026-04-01	April	Contoso Ltd	Enterprise	Widget Beta	Widgets	12	270.00
7	2026-04-15	April	Northwind Traders	Enterprise	Widget Alpha	Widgets	25	375.00
8	2026-05-02	May	Woodgrove Bank	Enterprise	Gadget Delta	Gadgets	6	360.00
9	2026-05-20	May	Fabrikam Inc	SMB	Component Epsilon	Components	40	399.60
10	2026-06-08	June	Woodgrove Bank	Enterprise	Widget Beta	Widgets	14	315.00
11	2026-06-15	June	Northwind Traders	Enterprise	Gadget Gamma	Gadgets	2	90.00



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Verify the final state

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Explore

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```
1  -- List all warehouse objects by schema and type
2  SELECT
3      TABLE_SCHEMA AS [schema],
4      TABLE_NAME AS [name],
5      TABLE_TYPE AS [type]
6  FROM INFORMATION_SCHEMA.TABLES
7  WHERE TABLE_SCHEMA IN ('staging', 'dim', 'fact', 'gold')
8  UNION ALL
9  SELECT
10     ROUTINE_SCHEMA,
11     ROUTINE_NAME,
12     ROUTINE_TYPE
13  FROM INFORMATION_SCHEMA.ROUTINES
14  WHERE ROUTINE_SCHEMA IN ('staging', 'dim', 'fact', 'gold')
15  ORDER BY [schema], [type], [name];
```

MessagesResults			
ABC	schema	name	type
1	dim	customer	BASE TABLE
2	dim	date	BASE TABLE
3	dim	product	BASE TABLE
4	fact	sales	BASE TABLE
5	gold	monthly_sales	BASE TABLE
6	gold	usp_refresh_monthly_sales	PROCEDURE
7	gold	vw_monthly_sales	VIEW
8	staging	customers	BASE TABLE
9	staging	dates	BASE TABLE
10	staging	orders	BASE TABLE
11	staging	products	BASE TABLE



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Reference

